```
{
    Dominant Cycle
    (C) 2026   John F. Ehlers
}

Inputs:
        LowerBound(10),
        UpperBound(40),
        WindowLength(27);

Vars:
        HP(0),
        LP(0),
        RMS(0),
        Real(0),
        ROC(0),
        QRMS(0),
        Imag(0),
        Angle(0),
        DC(0);

HP = $HighPass(Close, UpperBound);
LP = $SuperSmoother(HP, LowerBound);
RMS = $RMS(LP, 100);
If RMS <> 0 Then Real = LP / RMS;

ROC = Real - Real[1];
QRMS = $RMS(ROC, 100);
If QRMS <> 0 Then Imag = ROC / QRMS;

//Compute the angle as an arctangent function
If Real <> 0 Then Angle = 90 - Arctangent(Imag / Real);

//Resolve Arctangent ambiguity
If Real < 0 Then Angle = Angle - 180;

//compensate for filter lag / prediction
If Angle > 180 Then Angle = Angle - 360;

//angle cannot go backwards
If Angle < Angle[1] and Angle[1] - Angle < 270 Then Angle = Angle[1];

//Find DeltaAngle and eliminate outliers
If Angle <> Angle[1] Then DC = 360 / (Angle - Angle[1]);
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```
If DC > 50 Then DC = 50;
If DC < 8 Then DC = 8;

Plot1($Hann(DC, WindowLength));
```